

YaP MMO — Ability engine (M6)

Status: Shipped (M6)

Jars: `yap-abilities-api.jar`, `yap-abilities.jar`

Depends on: `yap-skills`, `yap-combat` (optional bridge), `yap-sched`

Overview

M6 adds a **config-driven ability catalog** separate from RS progression skills:

Concept	Count (v1)	Notes
RS progression skills	13	Unchanged (<code>yap-skills</code>)
Combat abilities	233	YAML packs in <code>plugins/YaPAilities/abilities/</code>
Status effects	14	Bufs/debuffs/DoTs in <code>effects/</code>

Abilities support:

- **Custom icons** on dedicated `CLAY_BALL` tokens (CustomModelData) — not swords/rods/tools
- **Cast VFX + sounds** (particles, volume/pitch)
- **Projectiles** (entity type, speed, trail particles, hit detection)
- **On-hit pipeline** (damage, knockback, buff/debuff apply, XP)
- **Costs** (prayer, runes, staff)
- **Cooldowns** and level gates per skill
- **Self / raycast** targeting + filters (`undead` , `player` , `mob`)

Folia-safe: all entity/world mutations via `YapSched` .

Commands

Command	Permission	Description
<code>/ability list [page]</code> <code>[category]</code>	<code>yapabilities.use</code>	Paginated ability list
<code>/ability cast <id></code>	<code>yapabilities.use</code>	Cast ability
<code>/ability info <id></code>	<code>yapabilities.use</code>	Show requirements
<code>/ability bar</code>	<code>yapabilities.use</code>	List hotbar bindings + current page
<code>/ability book [category]</code> <code>[page]</code>	<code>yapabilities.use</code>	Open ability book GUI

Command	Permission	Description
<code>/abilities</code>	<code>yapabilities.use</code>	Alias — open ability book
<code>/ability tome</code>	<code>yapabilities.use</code>	Receive an Ability Tome item
<code>/ability mode</code> <code>[build\ combat]</code>	<code>yapabilities.use</code>	Swap or set build/combat hotbar page
<code>/ability bind <1-6> [id]</code>	<code>yapabilities.bar</code>	Bind ability to bar slot (clears if id omitted)
<code>/ability clear</code>	<code>yapabilities.bar</code>	Clear all bar bindings
<code>/cast <id></code>	<code>combat</code>	Delegates to ability engine when loaded
<code>/yapabilities reload</code>	<code>yapabilities.admin</code>	Reload config + ability/effect YAML packs (hot)

Ability hotbar (dual page)

MMO **two-page hotbar** in `yap-abilities.jar`:

Page	Hotbar	Behavior
Build (default)	1-9	Normal Minecraft — blocks, tools, full row
Combat	1-3 weapons · 4-9 cast abilities	Press 4-9 to cast; weapon keys work normally

Swap build ↔ combat

Input	Config key
Middle mouse (pick block)	<code>PICK_BLOCK</code>
Swap hands (<code>F</code> — rebind to middle mouse in Minecraft Controls)	<code>SWAP_HANDS</code>
Sneak + Q (drop)	<code>SNEAK_DROP</code>
Command	<code>/ability mode</code> · <code>/ability mode build</code> · <code>/ability mode combat</code>

Config: `plugins/YaPAbilities/config.yml` → `ability-bar.*`

Bindings: `/ability bind 1 wind_strike ... /ability bar`

Persistence: `plugins/YaPAbilities/bars.yml` (ability ids; weapon rows stay in-session)

RS **skills** stay passive — only **abilities** bind to combat keys 4-9.

Ability book (spellbook GUI)

Browse abilities in a **chest GUI** (not a vanilla written book) and bind them to combat keys **4-9**.

Level-1 spells (Wind Strike, etc.) are bindable immediately. Higher spells show as gunpowder until the RS skill is high enough. Ops / `yapabilities.admin` can bind any ability.

Input	Action
<code>/abilities</code> <code>/ability</code> <code>/spell</code>	Open the ability book GUI
<code>/spell wind_strike</code>	Put that spell on the next empty hotbar key (4-9)
Hover a spell in the book	See what it does, costs, and requirements
Right-click a spell or <code>/ability info <id></code>	Full spell card in chat
Click a spell in the book	Add it to the hotbar and switch to the combat bar
Ability Tome (right-click enchanted book)	Open the book — auto-given on first join when enabled
<code>/ability tome</code>	Get a replacement tome
<code>/ability mode combat</code>	Show the combat hotbar, then press 4-9 to cast
Bedrock	Same commands open a form UI (category → ability → slot)

Java GUI

- **Category tabs** — All · Magic · Ranged · Melee · Prayer · Utility
- **Ability grid** — paginated; locked abilities shown gray when `show-locked: true`
- **Bottom row** — combat bar slots (keys 4-9)
- **Drag** an ability onto a slot, **shift-click** an ability for the first empty slot, **right-click** a slot to clear
- **Clear all** button removes every binding

Config: `plugins/YaPAbilities/config.yml` → `ability-book.*`
Persistence: same `bars.yml` as `/ability bind` — book and commands stay in sync.

Web dashboard

Gameplay → **MMO** tab (`GET /api/mmo`) shows ability catalog count, dual hotbar/book flags, and online combat bar bindings. Admins can hot-reload packs with **Reload abilities** (`POST {"action":"reload-abilities"}`). See [WEB_DASHBOARD.md](#).

Cast costs

By default only **skill level** gates casting. In `plugins/YaPAilities/config.yml` :

```
costs:
  require-runes: false
  require-staff: false
  require-prayer: false
```

Set any flag to `true` to enforce those costs again.

YAML schema (ability)

```
abilities:
  wind_strike:
    name: Wind Strike
    category: magic          # magic|ranged|melee|prayer|utility
    min-level:
      magic: 1
    costs:
      prayer: 1              # ignored unless costs.require-prayer=true
      runes:                 # ignored unless costs.require-runes=true
        FEATHER: 1
        LAPIS_LAZULI: 1
    cooldown: 0
    range: 20
    target: raycast          # raycast|self|none
    target-filter: undead    # optional
    cast:
      - type: vfx
        particle: CLOUD
        count: 10
      - type: sound
        sound: ENTITY_BREEZE_WIND_CHARGE_BURST
      - type: xp
        skill: magic
        amount: 12
        on-cast: true
    projectile:
      entity: SNOWBALL
      speed: 1.2
      max-ticks: 45
      trail:
        particle: CLOUD
        count: 3
        interval: 2
    on-hit:
      - type: damage
```

```
    style: magic          # magic|ranged|melee
    max-hit: 4
    - type: knockback
    power: 0.25
    - type: debuff
    id: poison
    stacks: 1
```

Effect types

Type	Purpose
damage	Combat roll via <code>AbilityCombatBridge</code> (yap-combat)
heal	Restore HP
vfx / sound	Cosmetics
buff / debuff	Apply status effect by id
knockback	Velocity push
xp	Grant skill XP (<code>on-cast: true</code> for cast phase)
teleport	Forward blink (<code>distance</code>)
velocity	Direct velocity set

Status effects

Defined in `effects/*.yaml` :

```
effects:
  poison:
    name: Poison
    kind: debuff
    duration-ticks: 100
    tick-interval: 20
    tick:
      - type: damage
        style: magic
        max-hit: 2
    modifiers:
      speed: 0.95
```

Prayer buff abilities reference matching effect ids (e.g. `protect_melee`).

Regenerating packs

```
python3 scripts/generate-ability-pack.py
```

Writes 233 abilities into `abilities-plugin/src/main/resources/abilities/`.

API

```
AbilityServices.find().ifPresent(s -> s.cast(player, "wind_strike"));
StatusEffectServices.find().ifPresent(s -> s.apply(target, "poison", sourceId, 1));
AbilityCombatServices.find(); // registered by yap-combat
```

M7 — Advanced mechanics + client graphics

Status: Shipped (M7)

Combat mechanics

Feature	YAML	Description
AoE	<code>on-hit: type: aoe</code> or <code>target: area</code>	Radius damage to filtered targets
Homing	<code>projectile.homing: true</code>	Steers toward locked/nearest target
Splash	<code>projectile.splash-radius: N</code>	AoE on projectile hit/expire
Chain	<code>on-hit: type: chain</code>	Jumps damage across nearby mobs
Conditions	<code>conditions: block</code>	Cast gates (HP%, status, mainhand, air/ground)
Animation	<code>cast: type: animation</code>	<code>swing</code> / <code>cast</code> / <code>channel</code> / <code>slam</code> + optional pose
Delay	<code>type: delay</code> + <code>ticks</code>	Sequence cast/hit effects across ticks

Example showcase abilities: `showcase_m7.yml` (`fireball_splash`, `homing_arc`, `chain_lightning`, ...).

Client graphics (cool cast FX)

Feature	How
Cast icon pulse	ItemDisplay + CustomModelData at caster eye (scales/floats ~14 ticks)
Spell projectile body	ItemDisplay rides the projectile; vanilla entity hidden (<code>projectile.hide: true</code>)

Feature	How
Particle shapes	shape: burst\ ring\ helix\ beam\ nova\ cone\ pillar\ orb\ shockwave on type: vfx
Dust / block data	color: R,G,B (DUST) · block: DIRT (BLOCK)
Sustained aura	ticks + interval on VFX
Non-blocking timed step	at: N on any effect — fires in N ticks without blocking the list (V1)
Shake	type: shake — velocity jitter pulses; optional radius for nearby players (V1)
Arc projectiles	projectile.path: arc + arc-height (V1)
Trail styles	trail.style: burst\ ribbon\ motion + falloff (V1)
Impact shake	projectile.impact-shake: true + shake-power (V1)
Impact burst	Auto nova + orb stage + sound when projectile lands
Hand sparkles	Crit/end-rod at hands on animation

Resource pack overlay: resourcepacks/yap-abilities/ (merged into yapcore-default.zip with -PyapGameplay=true).

Icon base item: CLAY_BALL (UI-only token — does not override swords, blaze rods, paper/vehicles, or rune items).

Range	Use
79000	Combat level (skills GUI)
79001-79013	RS skills
78010-78015	Showcase abilities
78200+	Generated abilities

Regenerate:

```
python3 scripts/generate-ability-pack.py
python3 scripts/generate-mmo-icons.py
```

```
cast:
  - type: animation
    style: cast          # swing|offhand|both|cast|channel|slam
    pulses: 2
    pose: glow          # glow|slow|levitate
  - type: vfx
```

```

particle: FLAME
shape: helix
count: 22
- type: vfx
  at: 4          # non-blocking – continues immediately
  particle: DUST
  shape: ring
  color: 255,80,20
- type: shake
  power: 0.08
  pulses: 2
projectile:
  entity: SNOWBALL
  path: arc      # straight|arc
  arc-height: 2.2
  hide: true
  scale: 0.95
  impact-shake: true
  trail:
    particle: FLAME
    count: 6
    interval: 1
    style: motion # burst|ribbon|motion
    falloff: 0.5

```

Example showcase abilities: `showcase_m7.yml` , `showcase_vfx.yml` (V1 arcs / timed layers / shake).

Bedrock spellbook

- MMO hub adds **Abilities** panel (paginated, tap-to-cast)
- `/mmoui abilities [category] [page]`

Roadmap (post-M7)

Ability VFX quality:  complete — [MMO_ABILITY_VFX.md](#) (V1–V4).

- **V1**  Engine: `at:` timelines, shake, arc paths, motion trails, cone/pillar/orb/shockwave
- **V2**  Element/archetype generator kits (`scripts/generate-ability-pack.py`)
- **V3**  Hand-authored heroes (`showcase_heroes.yml`)
- **V4**  Status ambient VFX, Bedrock ability-name cast bar, unique hero icons, Folia soak gates

Also still open (separate from VFX quality track):

- Broader per-spell resource pack art beyond heroes (all generated CMD models)
- Full Bedrock animation controller bridge (beyond action-bar cast labels)
- AoE shapes beyond circle
- Unify with combat-plugin `StatusEffectService` (single buff registry)